



▲ FAIRNESS IMPROVED

● NO ACCURACY LOSS

Executive Summary

Fairness Experiment Report - Upload 25 summarizes the current fairness state for upload 25. Accuracy changed to +0.923 and fairness score changed to +0.943. Strong fairness improvement with no measurable accuracy loss. The best balanced model is Pipeline_mitigated_smote with combined score 0.935.

ACCURACY	FAIRNESS SCORE	COMBINED SCORE	DISPARATE IMPACT
0.923 ▲ +0.923	0.943 ▲ +0.944	0.935 Pipeline_mitigated_smote	0.727 Ratio (DIR)

Dataset & Model Information

DATASET	MODEL	SECTIONS INCLUDED
processed_dataset.csv	german_dt_baseline.pkl	comparison

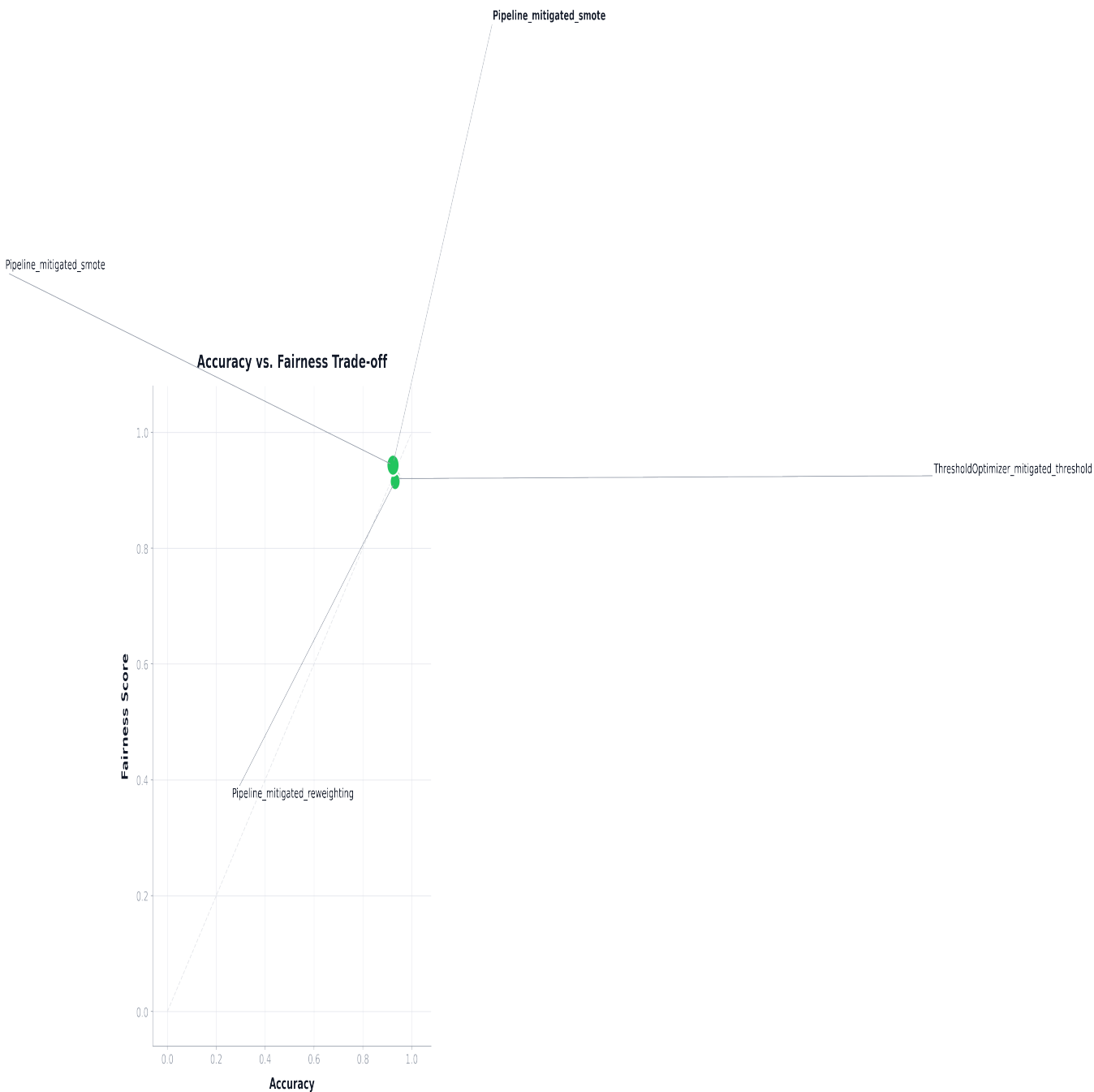
Fairness Metrics

METRIC	VALUE	INTERPRETATION
Demographic Parity Difference (DPD)	0.101	Difference in positive-prediction rates across groups; closer to 0 is more equitable.
Equalized Odds Difference (EOD)	0.012	Gap in true/false positive rates between groups; closer to 0 indicates equal error rates.
Disparate Impact Ratio (DIR)	0.727	Ratio of favorable outcome rates between groups; values near 1.0 indicate parity.

Model Comparison

Model	Accuracy	Fairness	DPD	EOD	DIR	Combined
★ Pipeline_mitigated_smote	0.923	0.943	0.101	0.012	0.727	0.935
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ThresholdOptimizer_mitigated_threshold	0.935	0.920	0.099	0.061	0.734	0.926
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ThresholdOptimizer_mitigated_threshold	0.935	0.920	0.099	0.061	0.734	0.926
Pipeline_mitigated_reweighting	0.932	0.915	0.100	0.069	0.722	0.922
Pipeline_mitigated_reweighting	0.932	0.915	0.100	0.069	0.722	0.922
Pipeline_mitigated_reweighting	0.932	0.915	0.100	0.069	0.722	0.922

Accuracy vs. Fairness Trade-off



Recommendations

Fairness improved after mitigation. Consider validating and deploying the mitigated model.